MARK FRANCALANGIA

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EDUCATION

Bachelor of Computer Science (BS), Southern New Hampshire University — October 2023
Software Engineering and Web Development (AS), New England Institute of Technology — September 2019
Achievements: 4.0 GPA, Honor Roll (2022–2023)

TECHNICAL SKILLS

Languages: C++, C#, Java, Python, HTML, CSS, JavaScript, Verse
Game Engines & Graphics: Unreal Engine, Unity, Panda3D, Three.js, WebGL
Backend & Architecture: Node.js, Express, WebSockets, REST APIs, MVC, Client-Server Networking, Scalable System Design
Databases: MySQL, PostgreSQL, Relational Schema Design
Infrastructure & Systems: vSphere, Ivanti, Hardware Integration, Vulnerability Management, System Deployment
Tools & Methods: Git, Docker, Vite, CI/CD, Agile/Scrum
Operating Systems: Linux, Windows

EXPERIENCE

IT Technician — *Systems Support, Hardware Integration, Networking, IT Support* — March 2025 - May 2026
Great Wolf Lodge - Mashantucket, Connecticut

- Provided technical support and troubleshooting for resort-wide systems, including POS/PMS, attraction hardware, and other guest-facing systems.
- Deployed and configured IT and hardware systems; performed break/fix, hardware repair, and managed tickets.
- Administered vSphere environments, assisted with on-site installations, and handled vulnerability management.
- Collaborated with operations and engineering teams to resolve system-level issues.

Volunteer Software Engineer — *Panda3D, Python, Astron, MMO Architecture* — March 2018 - Present
Toons of the World Foundation — Toontown Rewritten

- Designed and implemented client-server gameplay systems for a large-scale MMO supporting thousands of concurrent players.
- Contributed to distributed backend architecture using Astron and Panda3D, focusing on scalability, state synchronization, and performance.
- Integrated persistent data systems and relational database structures for player progression.
- Maintained and refactored legacy systems while introducing architectural improvements for long-term scalability.
- Collaborated with cross-functional teams in iterative Agile workflows to deliver stable production releases.

Software Engineer Intern — *Unity3D, C#, MVC Architecture* — June 2023 - August 2023
Indie.io

- Developed and maintained gameplay systems and UI components using Unity3D and C# following MVC architectural principles.
- Implemented scalable feature updates and optimized runtime performance for post-launch releases.
- Collaborated with engineers, designers, and QA in an Agile development environment to deliver stable builds.
- Contributed to debugging, refactoring, and architectural improvements within existing codebases.

PROJECTS

Toon Tagger — *JavaScript, Three.js, Node.js, Express, WebSockets, Vite*

- Recreated EPCOT's classic Toon Tag attraction as a browser-based 3D multiplayer game using Three.js and WebGL, playable without downloads or signup.
- Built matchmaking and room systems with Node.js and WebSockets, supporting public matches and private games through room codes.
- Implemented real-time tag gameplay with online multiplayer, practice against bots, and keyboard and touch controls.